Oxidative Phosphorylation



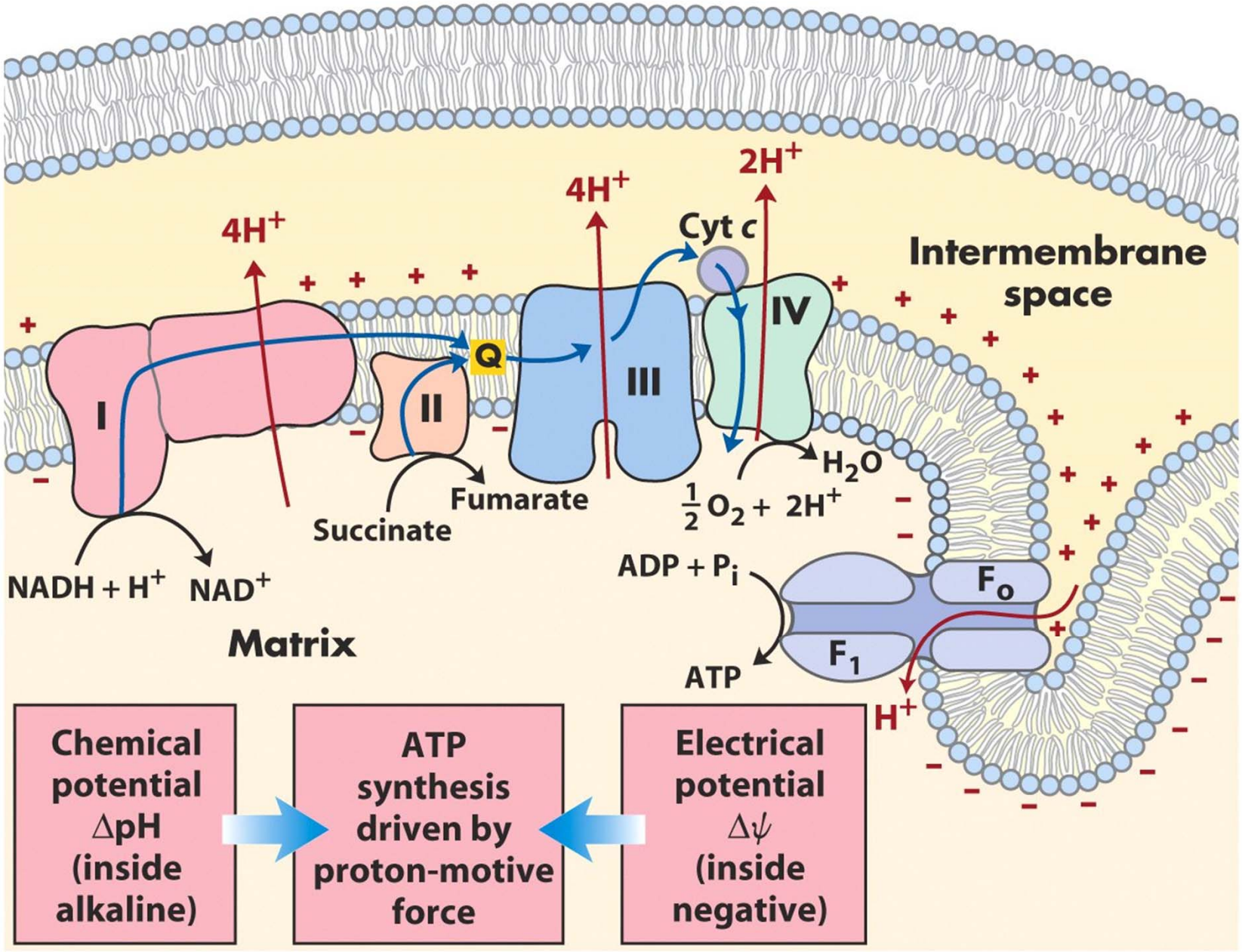
Energy from Reduced Fuels Is Used to Synthesize ATP in Animals

- Carbohydrates, lipids, and amino acids are the main reduced fuels for the cell.
- Electrons from reduced fuels are transferred to reduced cofactors NADH or $FADH_2$.
- In oxidative phosphorylation, energy from NADH and $FADH_2$ is used to make ATP.

Chemiosmotic Theory

- ADP + P_i → ATP is highly thermodynamically unfavorable.
- How do we make it possible?
- Phosphorylation of ADP is not a result of a direct reaction between ADP and some high-energy phosphate carrier.
- Energy needed to phosphorylate ADP is provided by the flow of protons down the electrochemical gradient.
- The energy released by electron transport is used to transport protons against the electrochemical gradient.

Chemiosmotic Theory



Chemiosmotic Energy Coupling Requires Membranes

- The proton gradient needed for ATP synthesis can be stably established across a membrane that is impermeable to ions.
  - inner membrane in mitochondria
- The membrane must contain proteins that couple the "downhill" flow of electrons in the electron-transfer chain with the "uphill" flow of protons across the membrane.
- The membrane must contain a protein that couples the "downhill" flow of protons to the phosphorylation of ADP.

Structure of a Mitochondrion

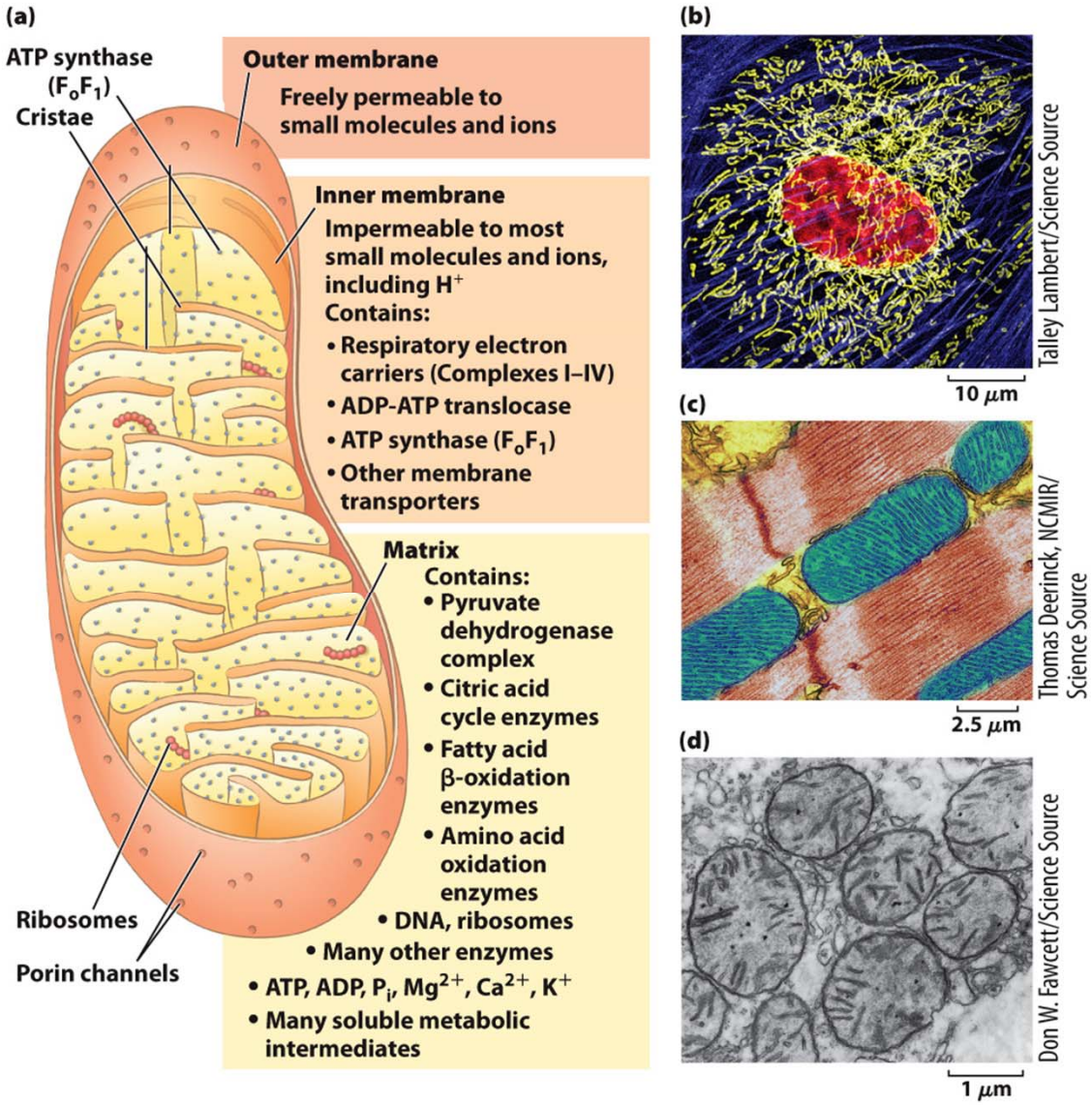


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Structure of a Mitochondrion

Double membrane leads to four distinct compartments:

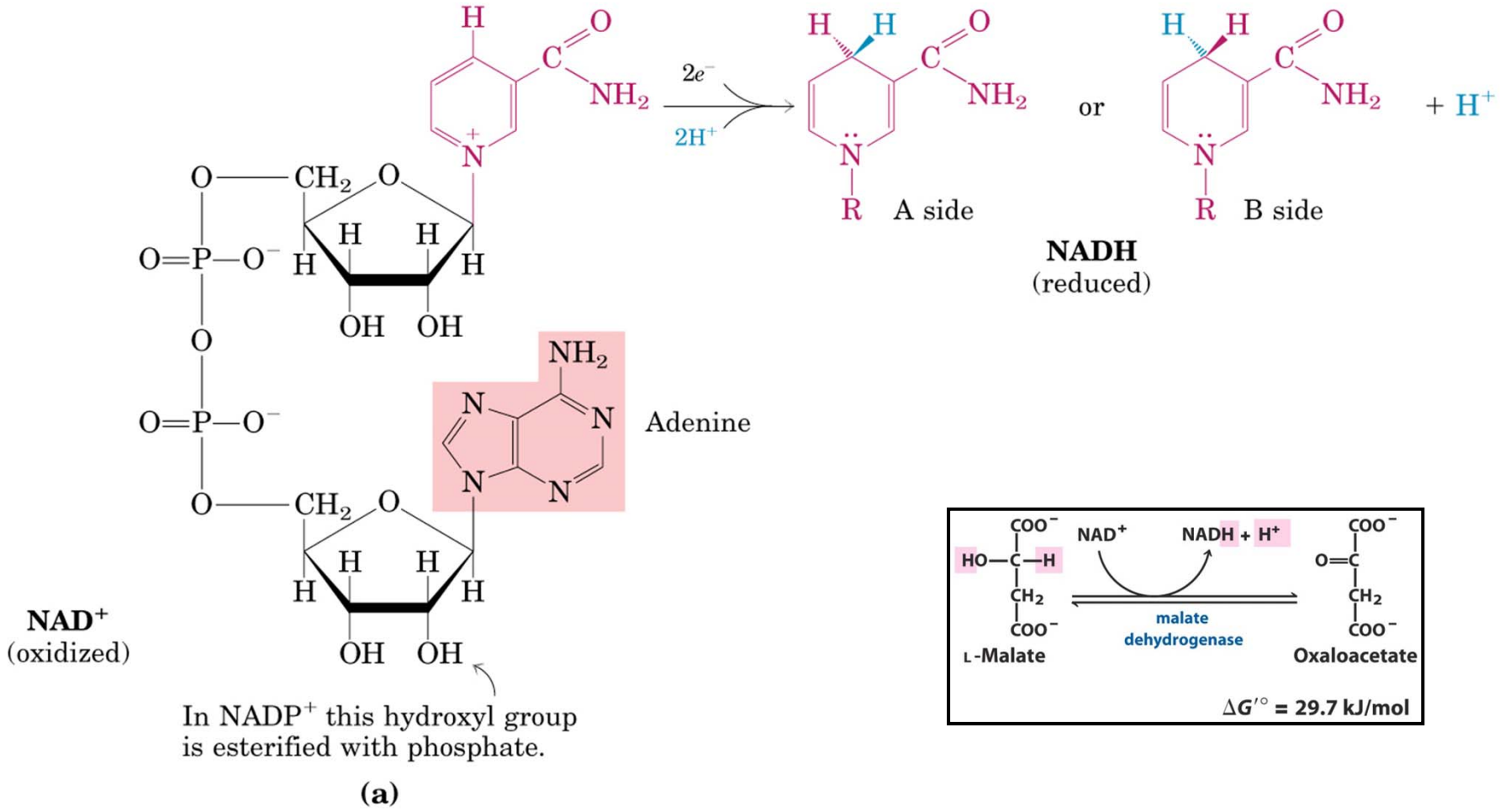
1. Outer membrane:
   - relatively porous membrane; allows passage of metabolites
2. Intermembrane space (IMS):
   - similar environment to cytosol
   - higher proton concentration (lower pH)
3. Inner membrane
   - relatively impermeable, with proton gradient across it
   - location of electron transport chain complexes
   - Convolutions called cristae serve to increase the surface area.
4. Matrix
   - location of the citric acid cycle and parts of lipid and amino acid metabolism
   - lower proton concentration (higher pH)

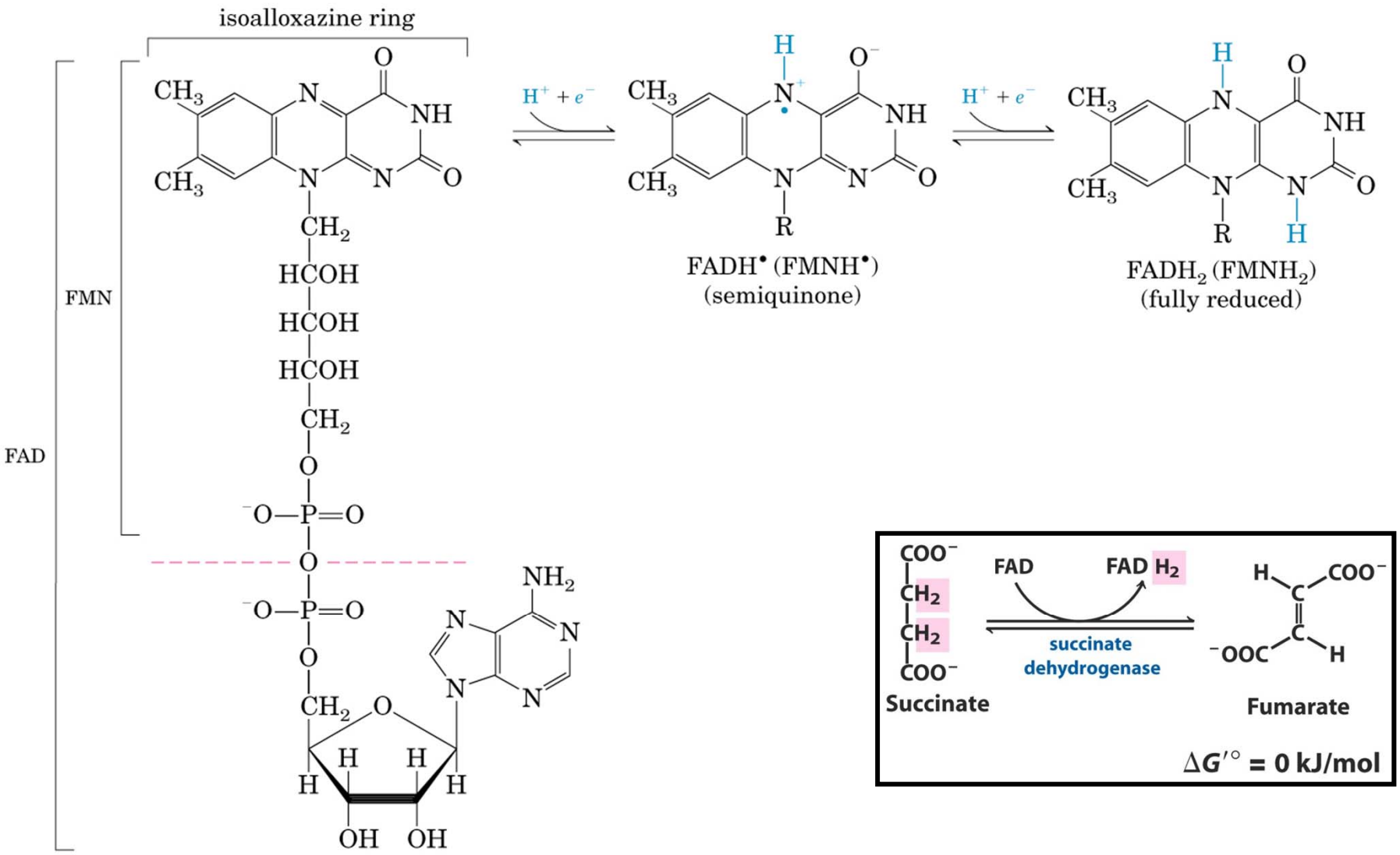
Electron-Transport Chain Complexes Contain a Series of Electron Carriers

- Each complex contains multiple redox centers consisting of:
  - flavin mononucleotide (FMN) or flavin adenine dinucleotide (FAD)
  - cytochromes *a*, *b*, or *c*
  - iron-sulfur cluster
- Order of transfer of electrons is dependent on reduction potential:

TABLE 19–2 Standard Reduction Potentials of Respiratory Chain and Related Electron Carriers

Redox reaction (half-reaction)	E'° *(V)*
$2H^+ + 2e^- \longrightarrow H_2$	−0.414
$NAD^+ + H^+ + 2e^- \longrightarrow NADH$	−0.320
$NADP^+ + H^+ + 2e^- \longrightarrow NADPH$	−0.324
NADH dehydrogenase (FMN) + $2H^+ + 2e^- \longrightarrow$ NADH dehydrogenase ($FMNH_2$)	−0.30
Ubiquinone + $2H^+ + 2e^- \longrightarrow$ ubiquinol	0.045
Cytochrome *b* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *b* (Fe^{2+})	0.077
Cytochrome c_1 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome c_1 (Fe^{2+})	0.22
Cytochrome *c* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *c* (Fe^{2+})	0.254
Cytochrome *a* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *a* (Fe^{2+})	0.29
Cytochrome a_3 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome a_3 (Fe^{2+})	0.35
$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O$	0.8166

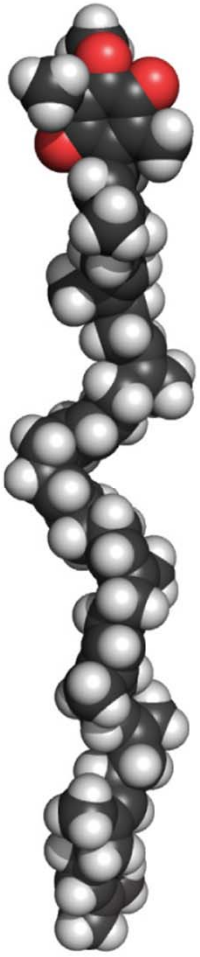

2e⁻
2H⁺
A side
or
B side
+ H⁺
NADH
(reduced)
Adenine
NAD⁺
(oxidized)
In NADP⁺ this hydroxyl group
is esterified with phosphate.
(a)
NAD⁺
NADH + H⁺
malate
dehydrogenase
L-Malate
Oxaloacetate
ΔG′° = 29.7 kJ/mol


isoalloxazine ring
FMN
FAD
$H^+ + e^-$
$H^+ + e^-$
FADH• (FMNH•)
(semiquinone)
$FADH_2$ ($FMNH_2$)
(fully reduced)
Flavin adenine dinucleotide (FAD) and
flavin mononucleotide (FMN)
FAD
$FAD H_2$
succinate
dehydrogenase
Succinate
Fumarate
$\Delta G'^{\circ}$ = 0 kJ/mol

Coenzyme Q or Ubiquinone

- Ubiquinone is a lipid-soluble conjugated dicarbonyl compound that readily accepts electrons.
- Upon accepting two electrons, it picks up two protons to give an alcohol, ubiquinol.
- Ubiquinol can freely diffuse in the membrane, carrying electrons with protons from one side of the membrane to another side.
- Coenzyme Q is a mobile electron carrier transporting electrons from Complexes I and II to Complex III.

Coenzyme Q or Ubiquinone



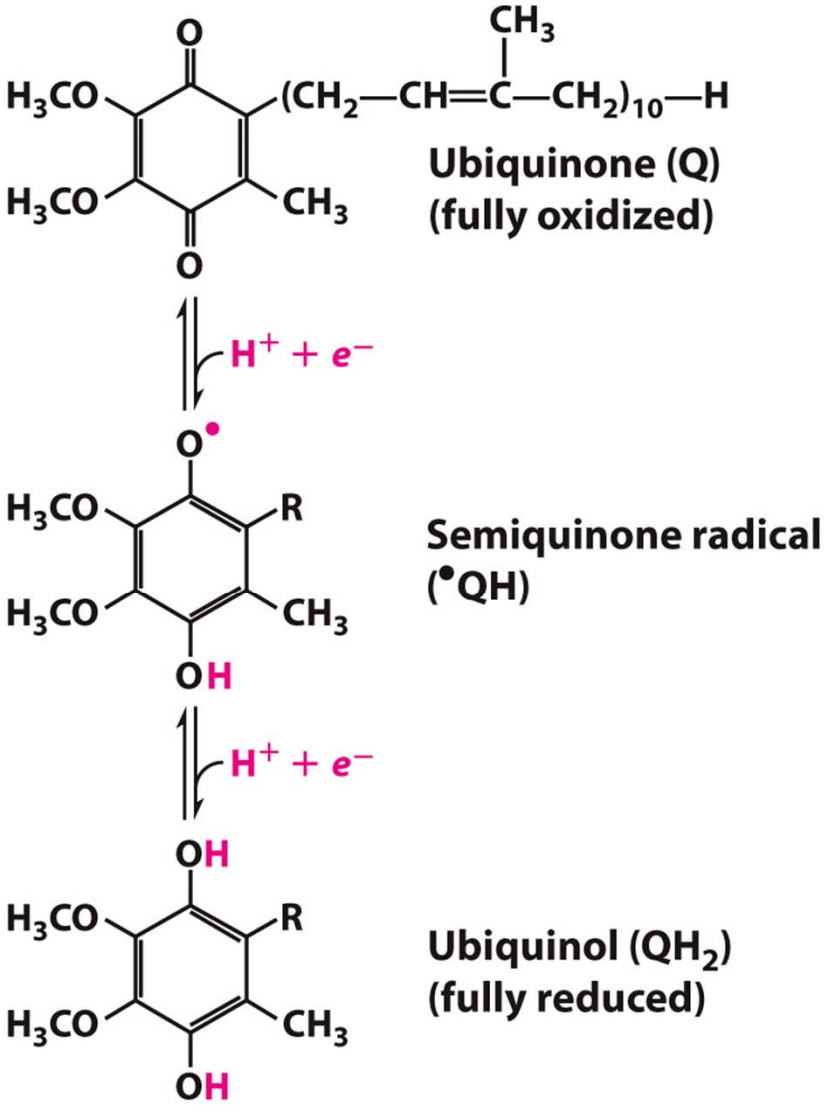


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Iron-Sulfur Clusters

- One-electron carriers
- Coordinating by cysteines in the protein
- Containing equal number of iron and sulfur atoms

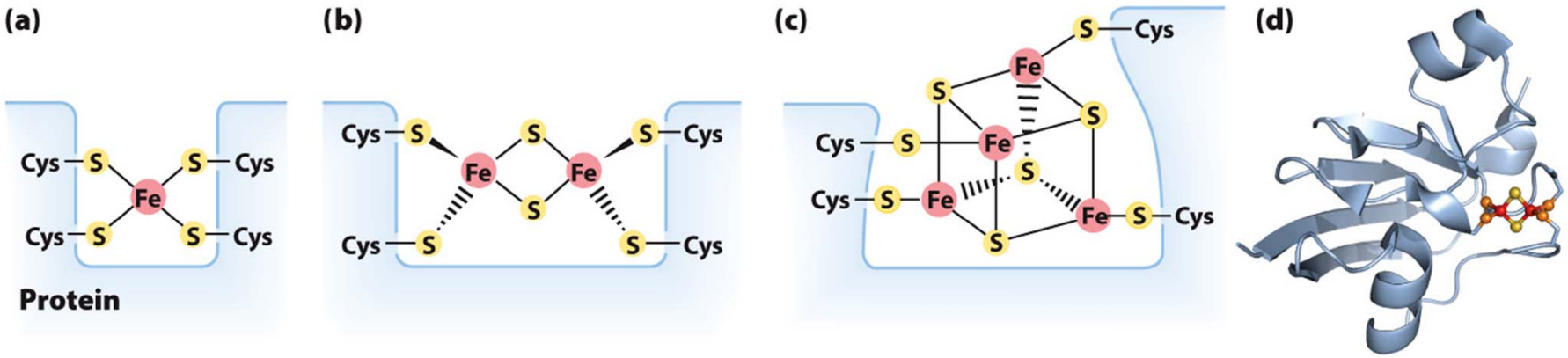
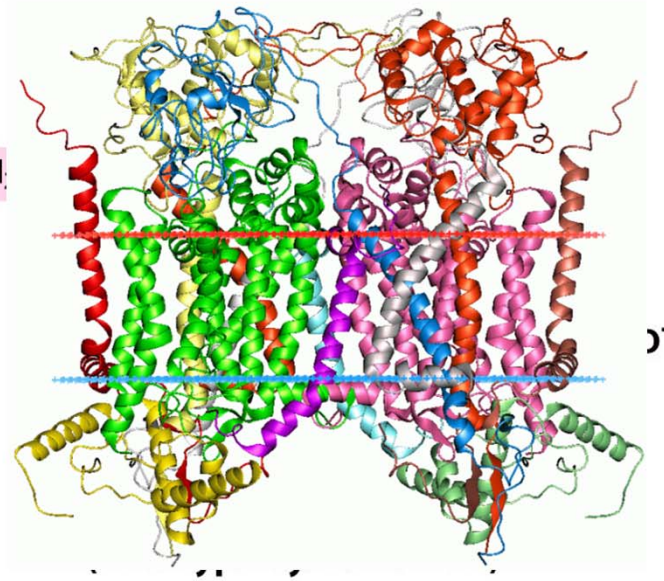
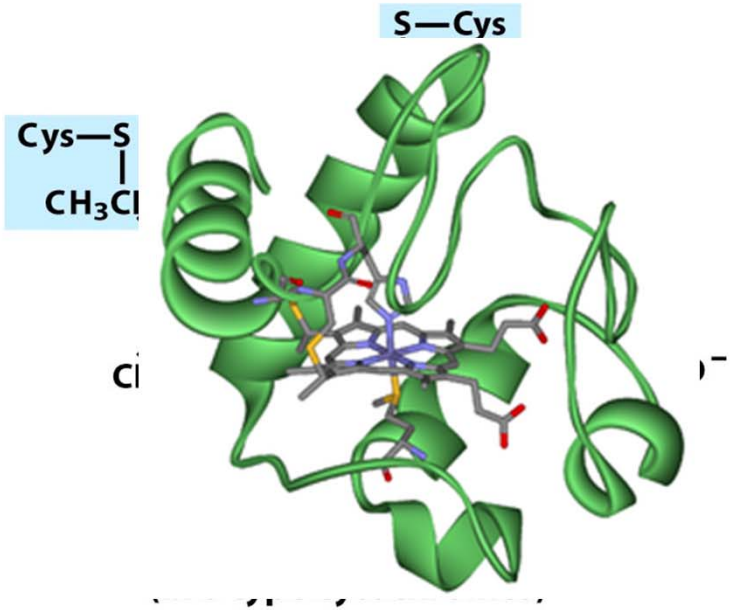


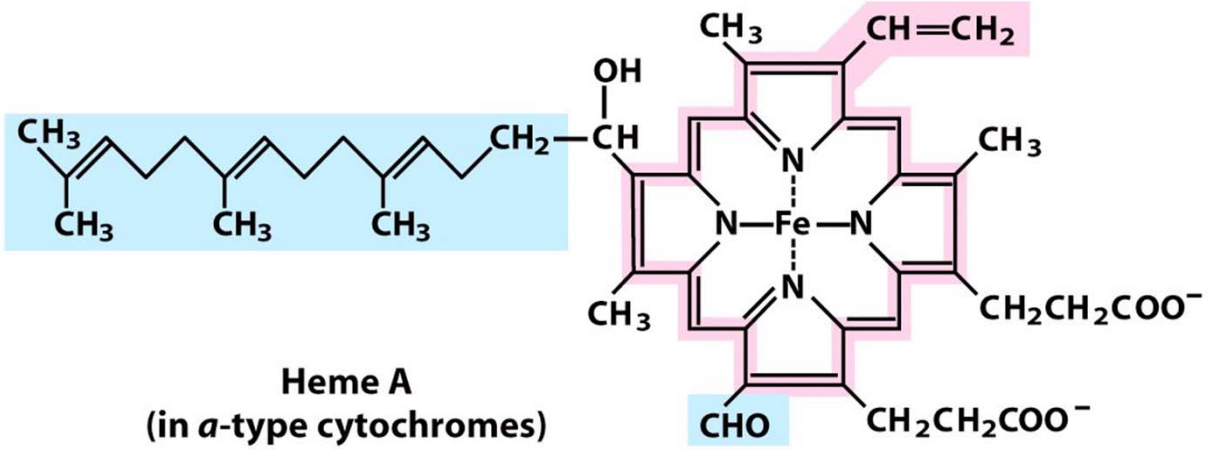
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Cytochromes

- One-electron carriers
- Iron coordinating porphoryin ring derivatives
- *a*, *b*, or *c* differ by ring additions







Heme A
(in *a*-type cytochromes)

Electron transporters are grouped in four complexes

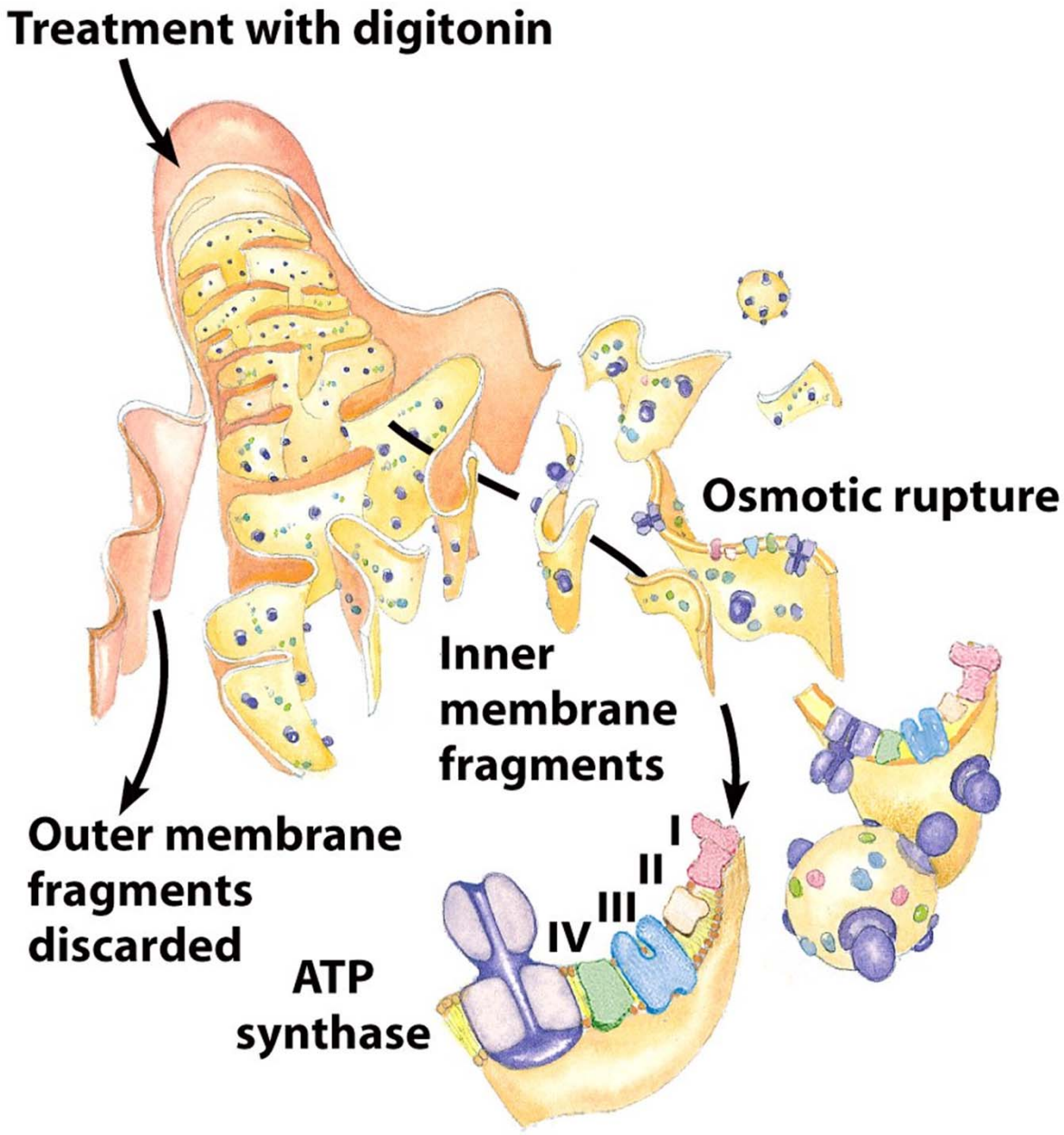
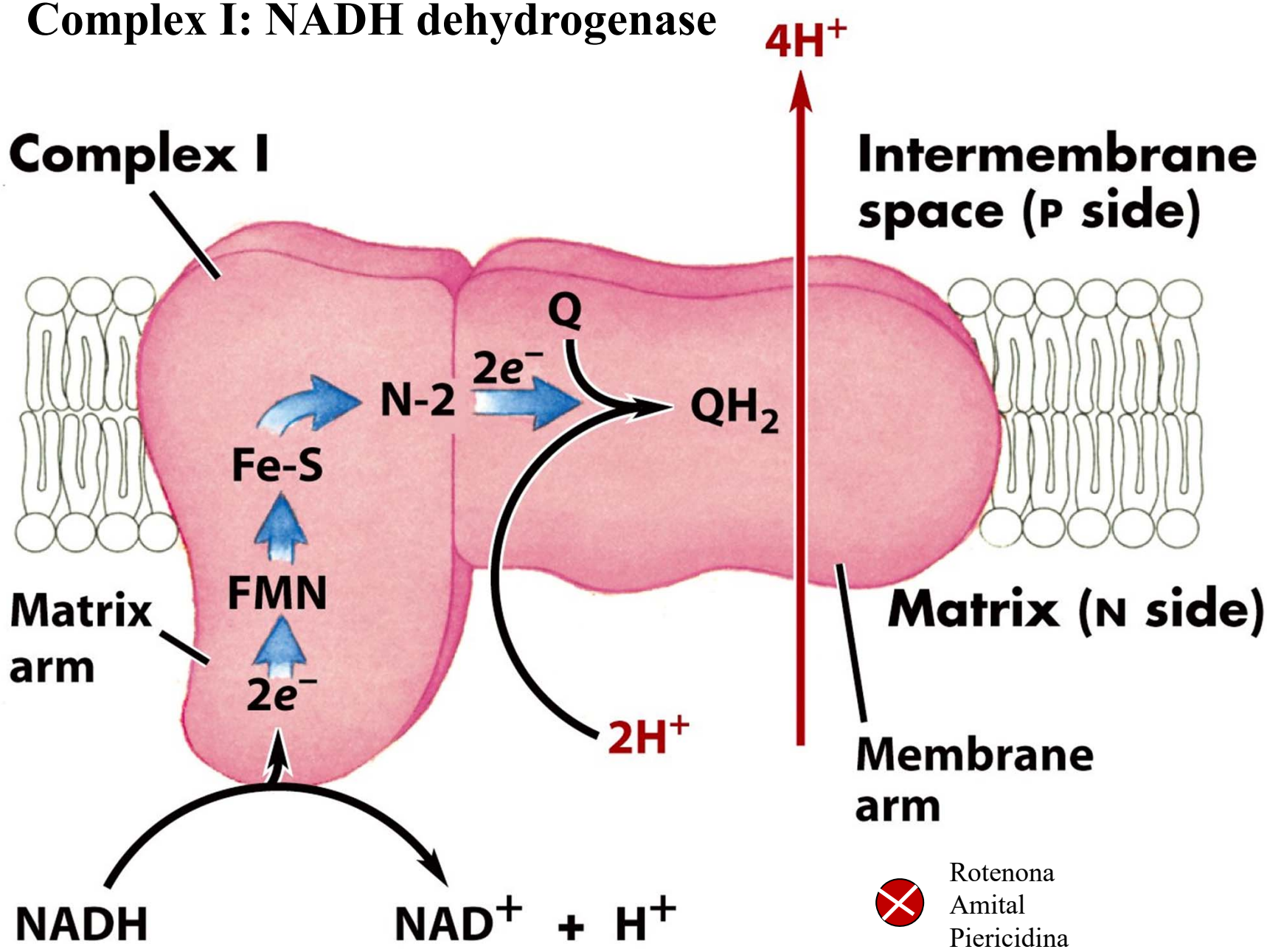


TABLE 19–3 The Protein Components of the Mitochondrial Electron-Transfer Chain

	Enzyme complex/protein	*Mass (kDa)*	*Number of subunits*[*]	*Prosthetic group(s)*
I	NADH dehydrogenase	850	43 (14)	FMN, Fe-S
II	Succinate dehydrogenase	140	4	FAD, Fe-S
III	Ubiquinone cytochrome *c* oxidoreductase	250	11	Hemes, Fe-S
	Cytochrome *c*[†]	13	1	Heme
IV	Cytochrome oxidase	160	13 (3–4)	Hemes; Cu_A, Cu_B

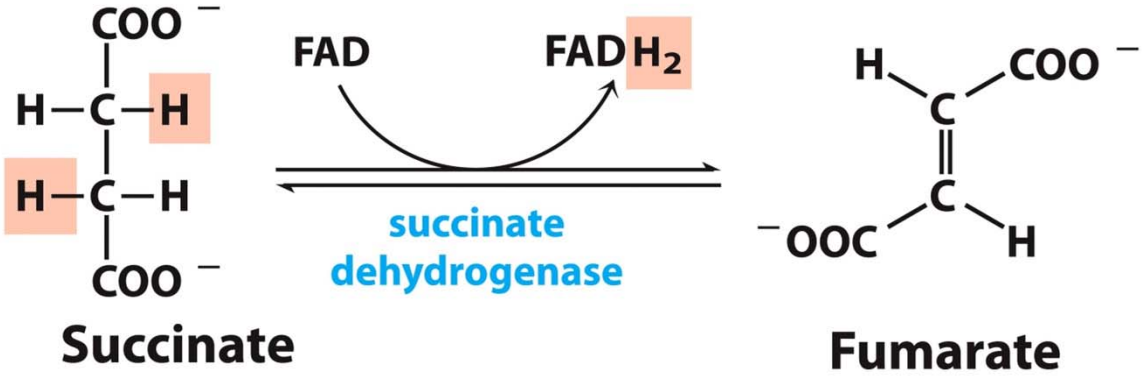
[*]Numbers of subunits in the bacterial equivalents in parentheses.
[†]Cytochrome *c* is not part of an enzyme complex; it moves between Complexes III and IV as a freely soluble protein.

Complex I: NADH dehydrogenase



Succinate Dehydrogenase, a.k.a. Complex II

- FAD accepts two electrons from succinate.
- Electrons are passed, one at a time, via iron-sulfur centers to ubiquinone, which becomes reduced QH_2.
- Does not transport protons



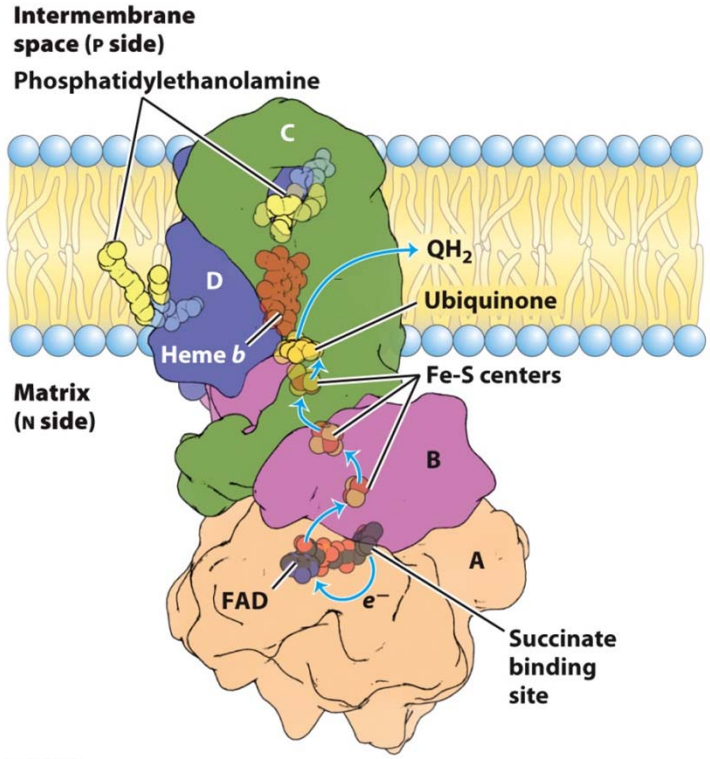
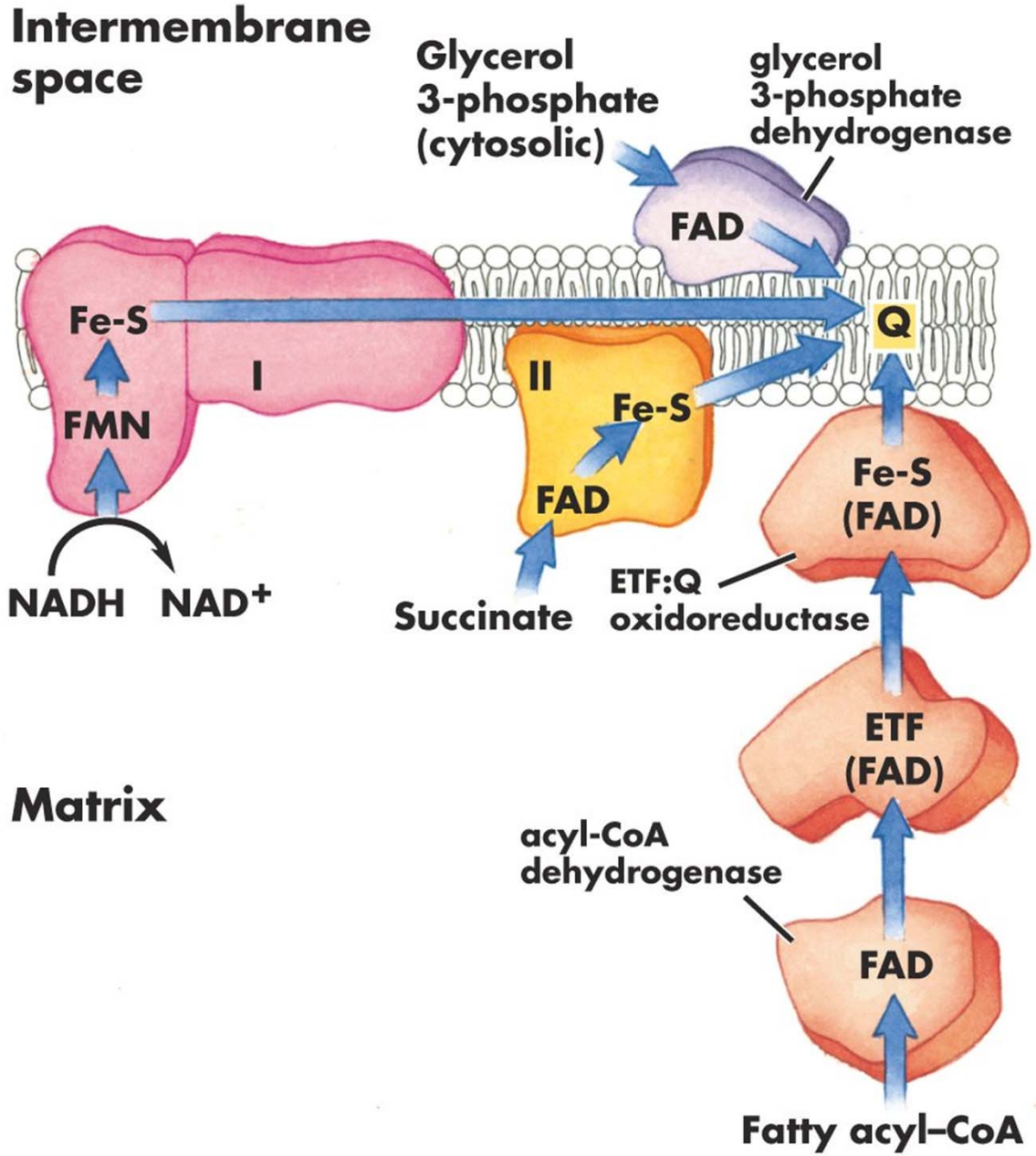


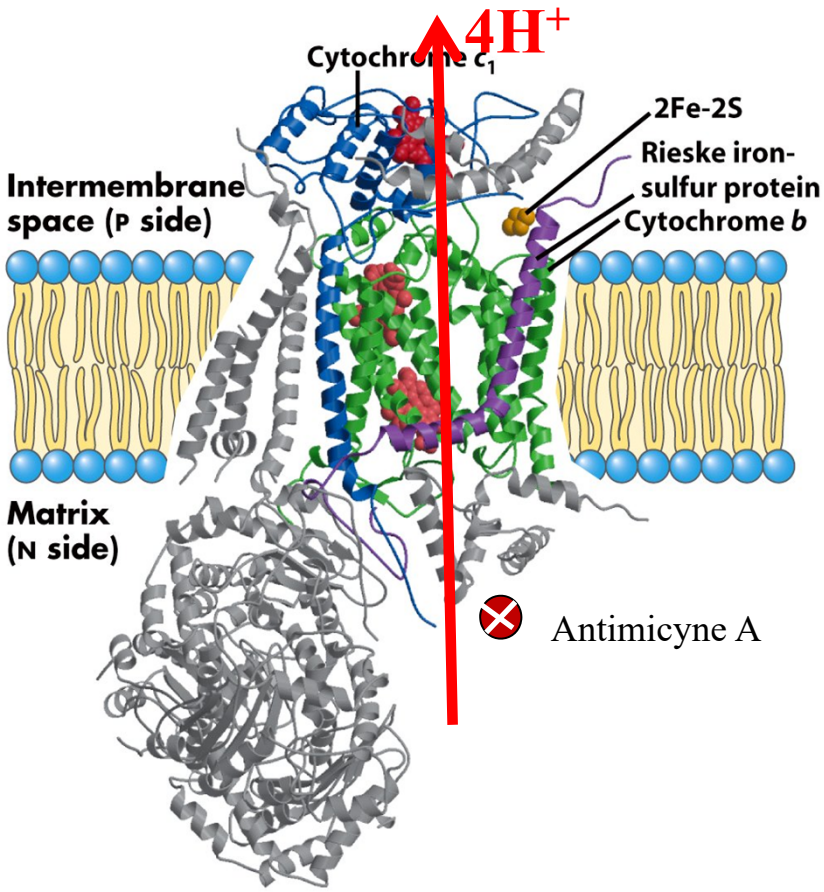
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The entrance gates for electrons into the Phosfox

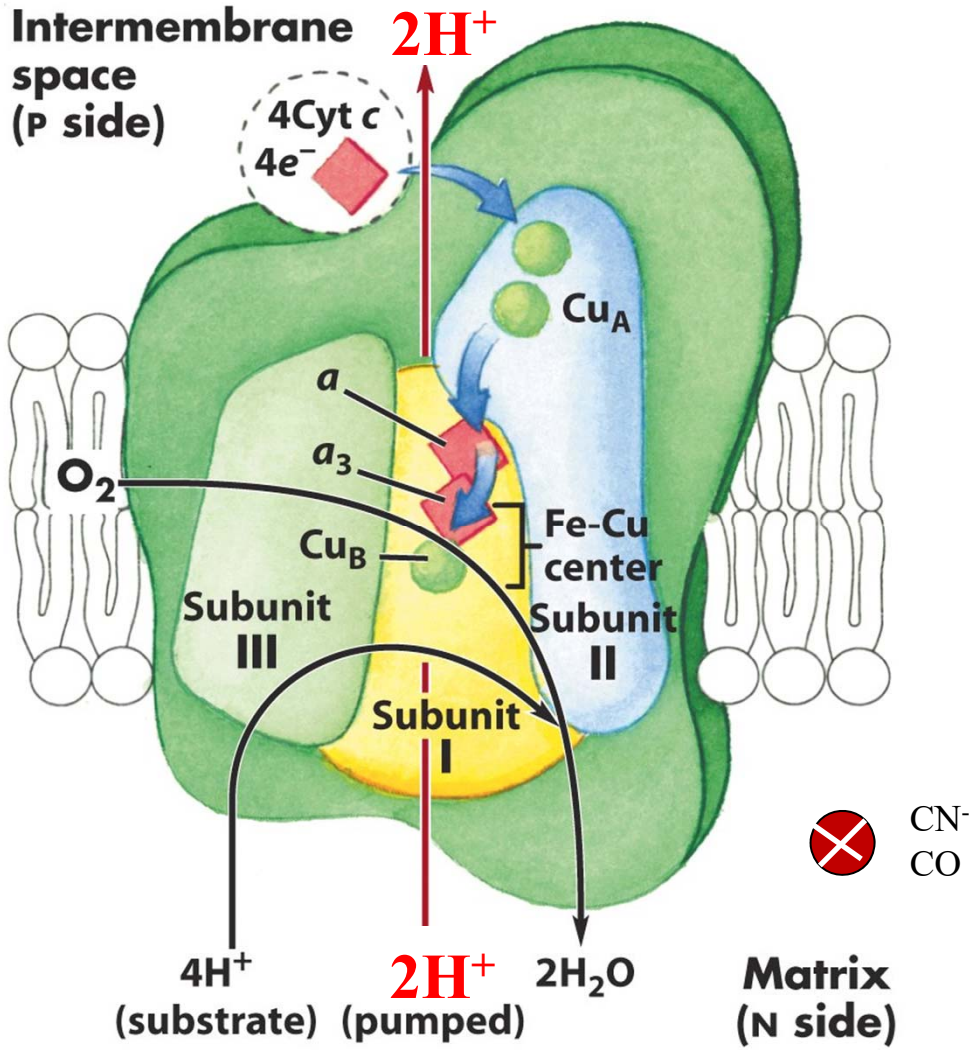


Ubiquinone:Cytochrome *c* Oxidoreductase, a.k.a. Complex III

- Uses two electrons from QH_2 to reduce two molecules of cytochrome *c*
- Additionally contains iron-sulfur clusters, cytochrome *b*, and cytochrome *c*
- Clearance of electrons from the reduced quinones via the Q-cycle results in translocation of four additional protons to the intermembrane space.



Complex IV: Cytochrome c oxidase



Summary of the Electron Flow in the Respiratory Chain

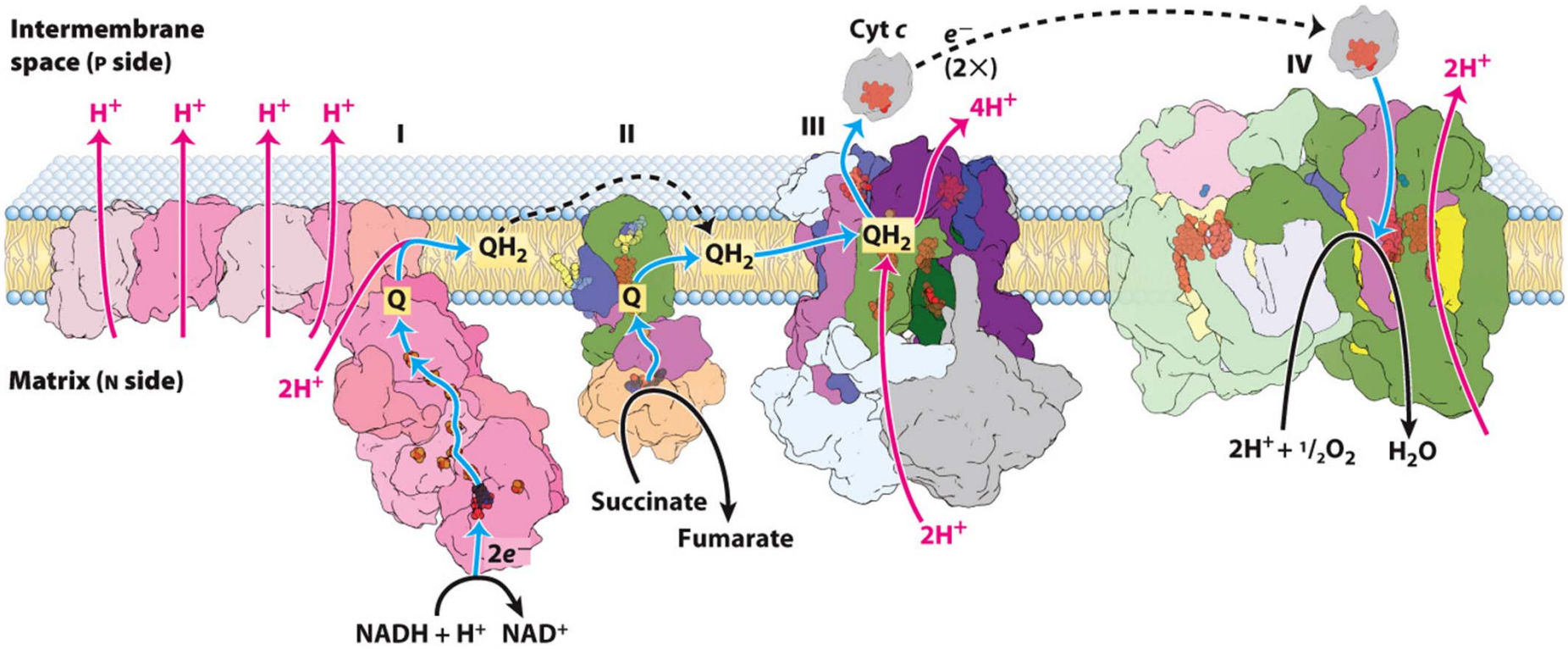


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Reactive Oxygen Species Can Damage Biological Macromolecules

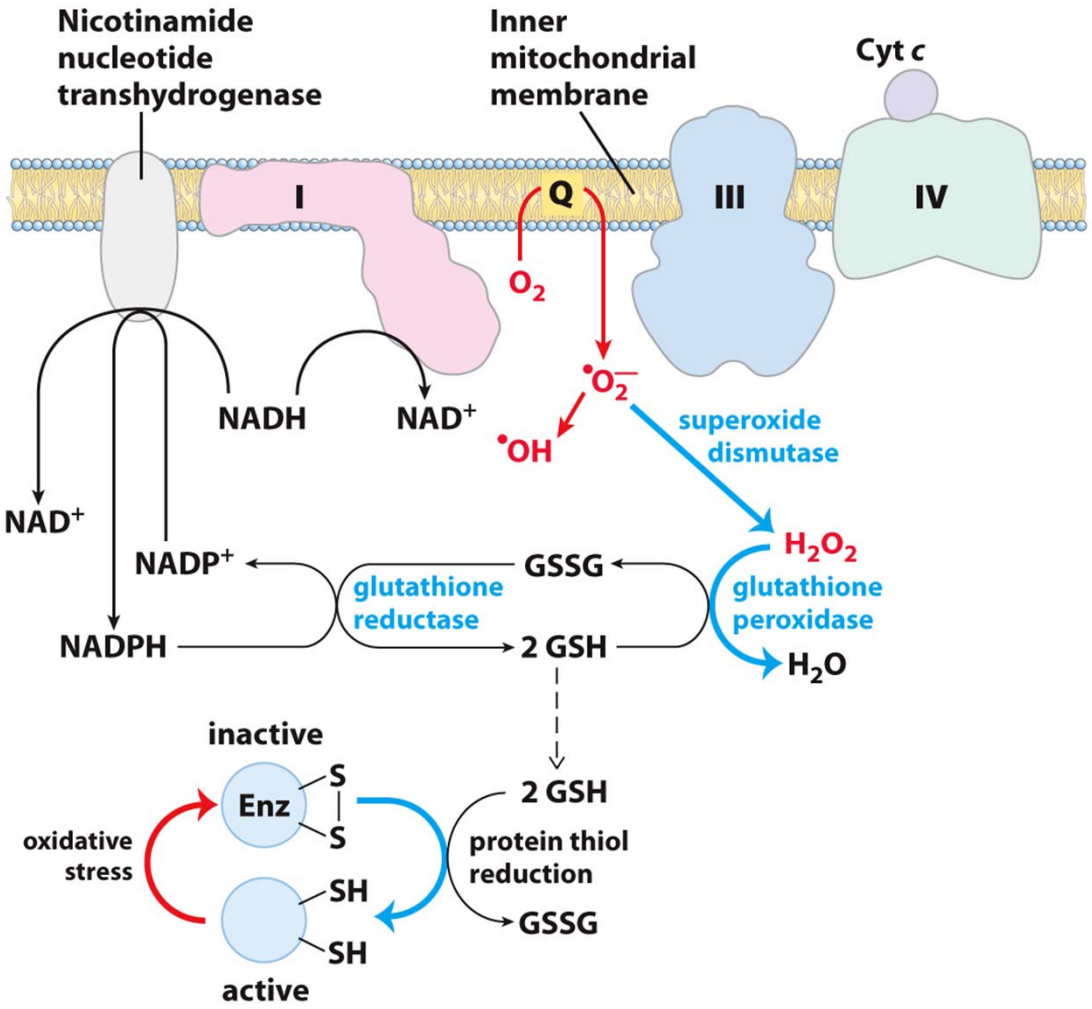


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- Ubiquinone is naturally "leaky" and facilitates partial reduction of non-Complex III targets.
  - Single electron tranfers result in free radicals.
- One method by which the cell can correct free-radical production of reduced glutathione, which fuels the glutathione shuttle

Chemiosmotic Model for ATP Synthesis

- Electron transport sets up a proton-motive force.
- Energy of proton-motive force drives synthesis of ATP.

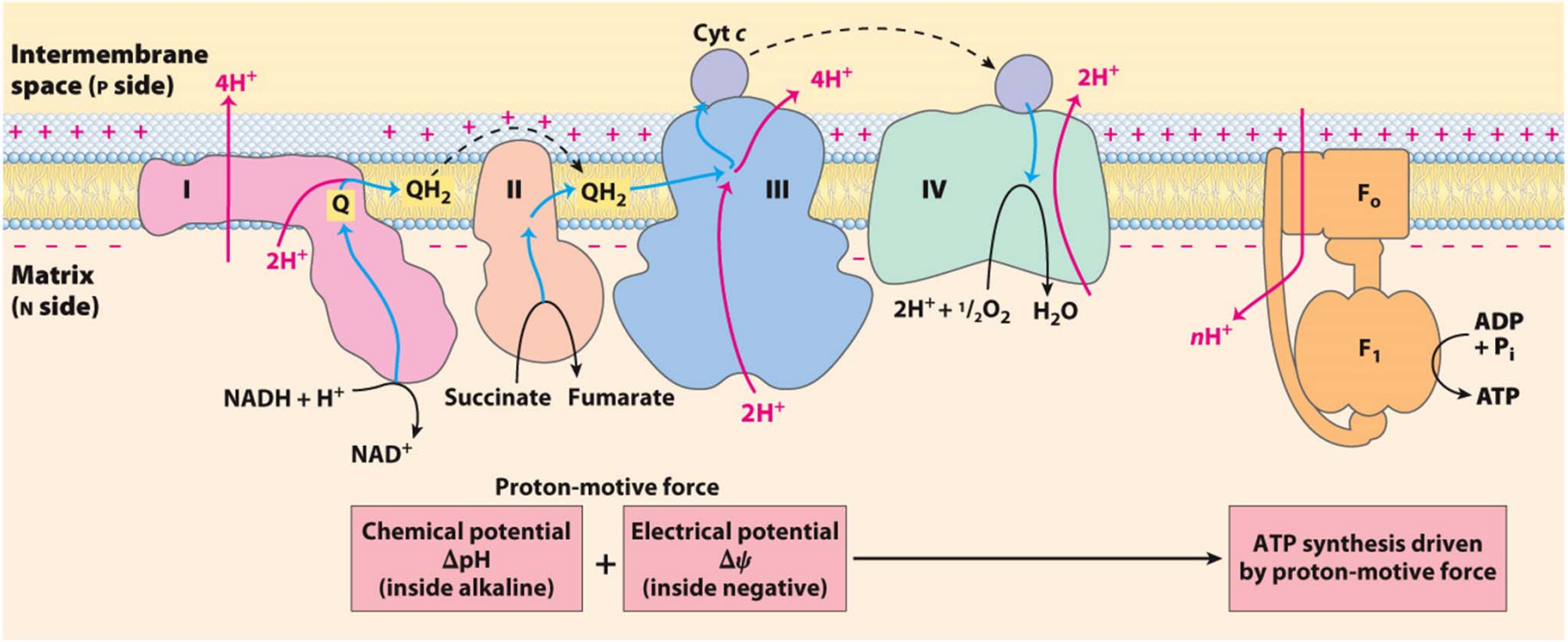
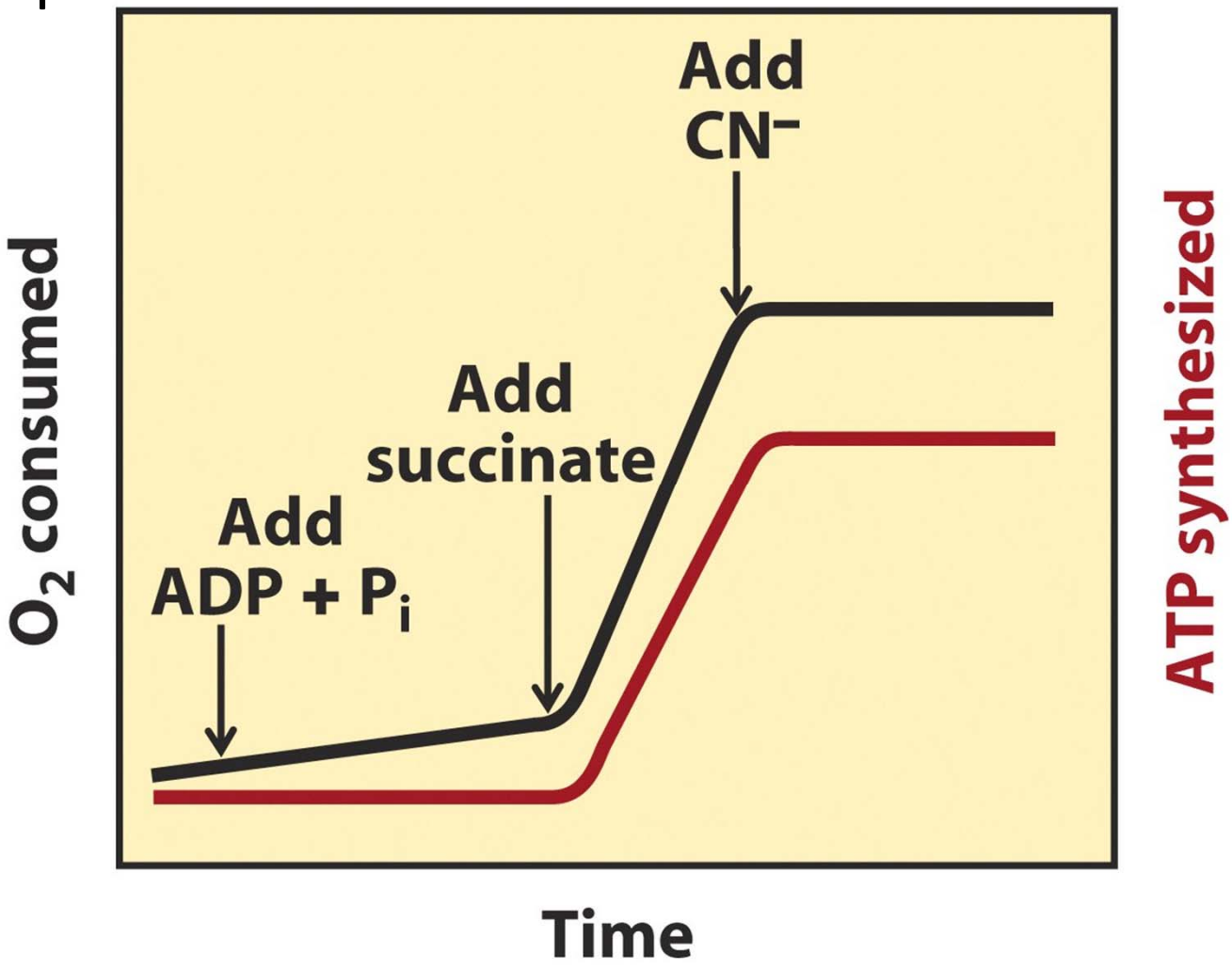


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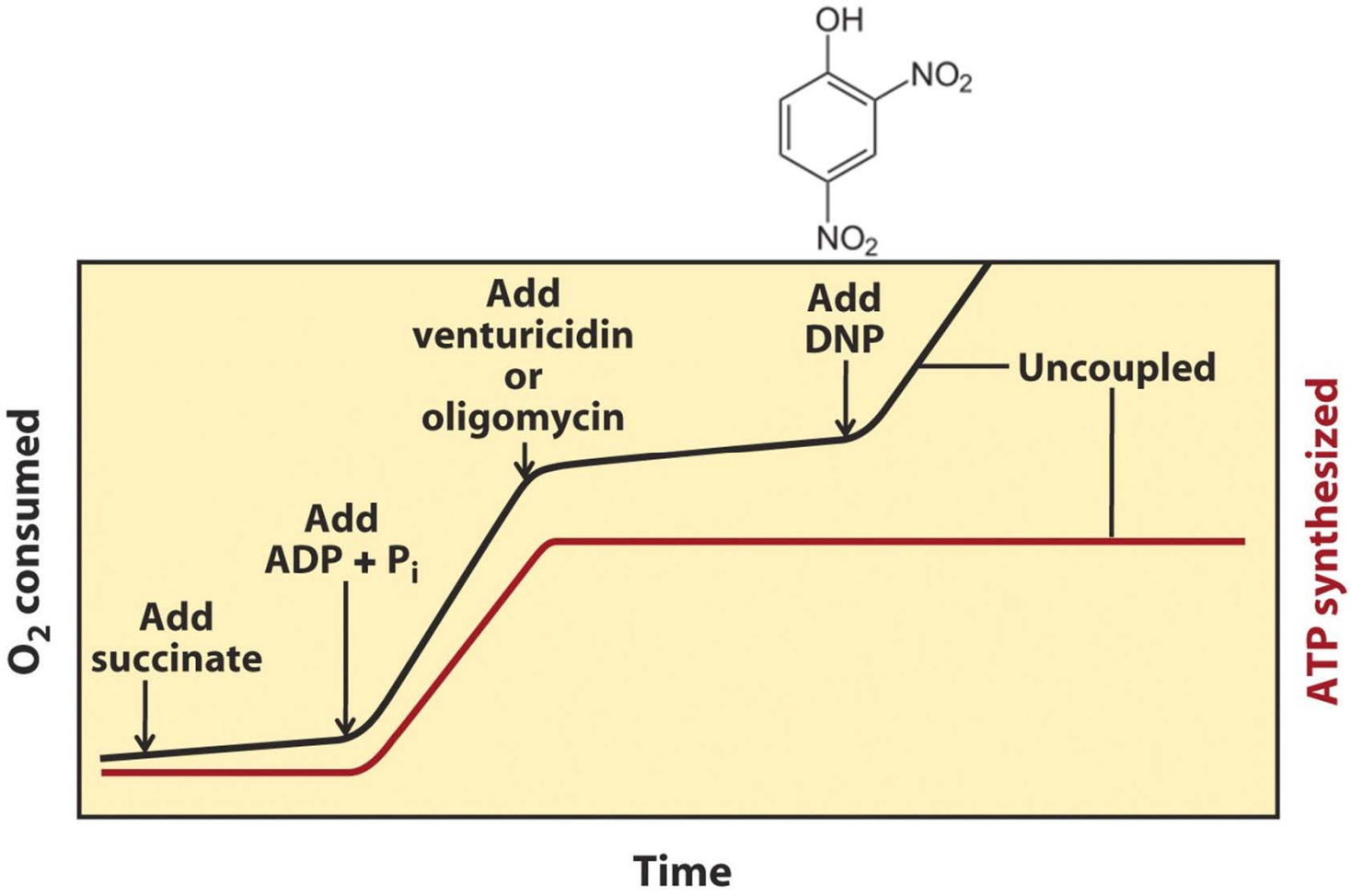
Relationship of ETC and ATP Synthesis<

- As described, ATP synthesis requires electron transport.

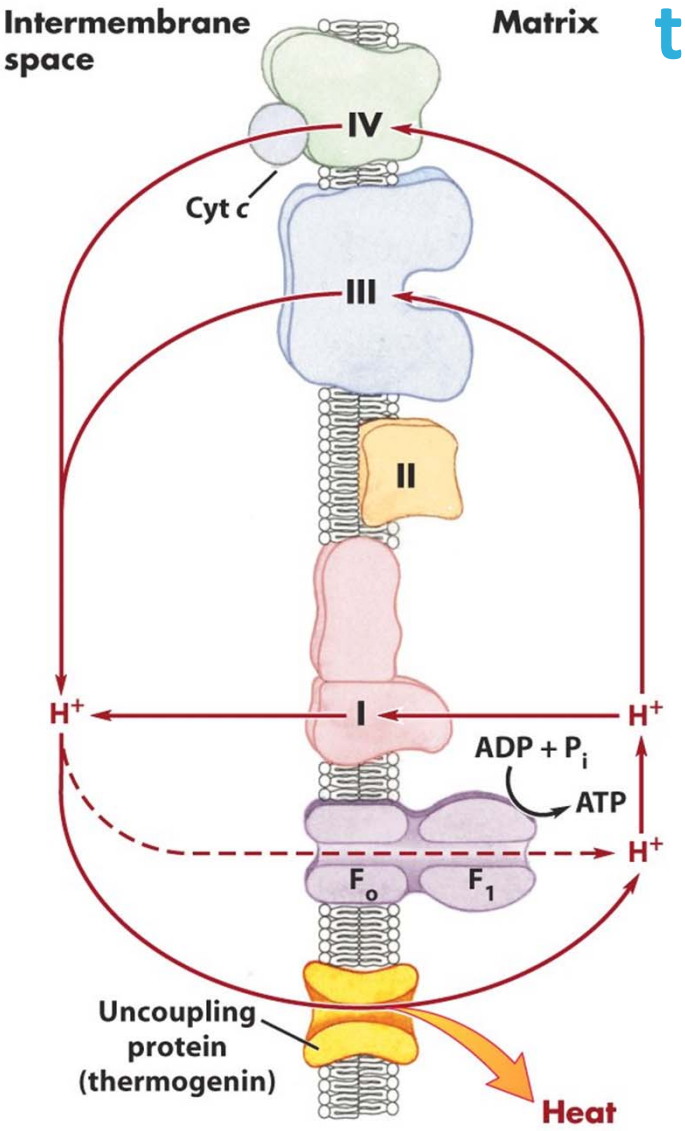


Relationship of ETC and ATP Synthesis<

- But electron transport does not requires ATP synthesis.



ATP Synthesis Can Be Uncoupled from ETC to Generate Heat



- **Thermogenin (UCP)** is an example of the physiological use of uncoupling to generate heat
- Brown adipose tissue mitochondria contains high amount of UCP
- UCP is activated by fatty acids

Mitochondrial ATP Synthase Complex

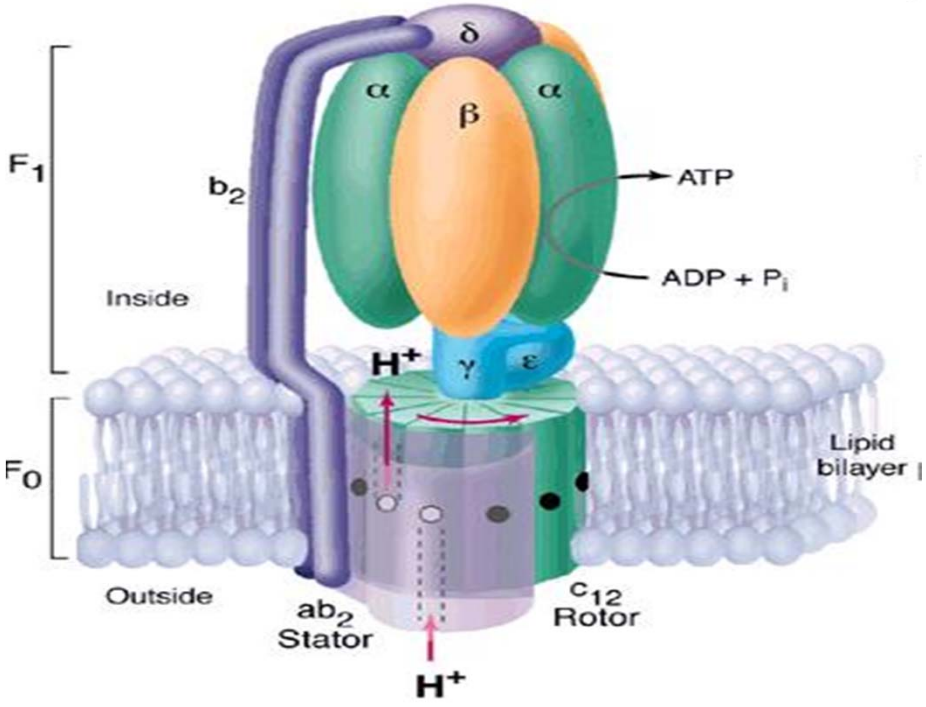
Contains two functional units:

F_1

- soluble complex in the matrix
- individually catalyzes the hydrolysis of ATP

F_0

- integral membrane complex
- transports protons from IMS to matrix, dissipating the proton gradient
- energy transferred to F_1 to catalyze phosphorylation of ADP



The F_1 catalyzes ADP + $P_i \rightleftharpoons$ ATP

- Hexamer arranged in three $\alpha\beta$ dimers
- Dimers can exist in three different conformations:
  - open: empty
  - loose: binding ADP and P_i
  - tight: catalyzes ATP formation and binds product

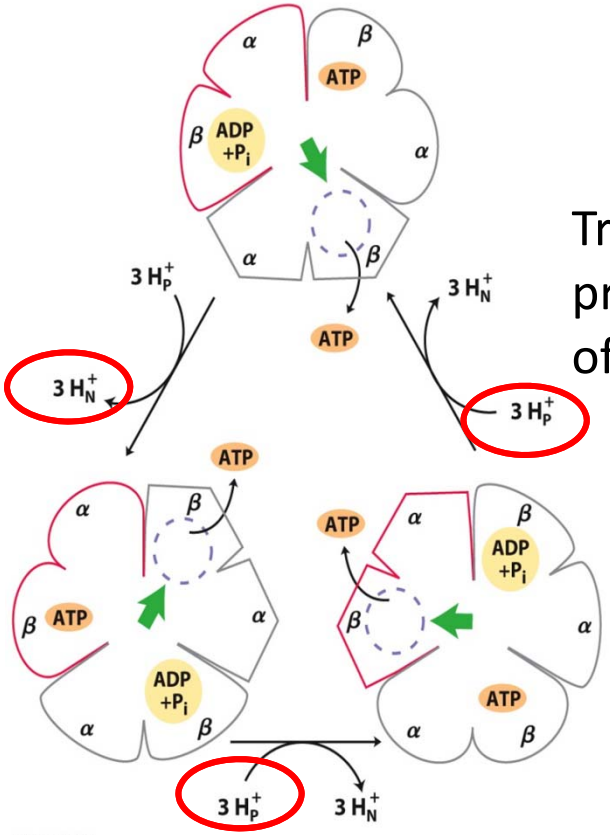


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Translocation of three protons fuels synthesis of one ATP.

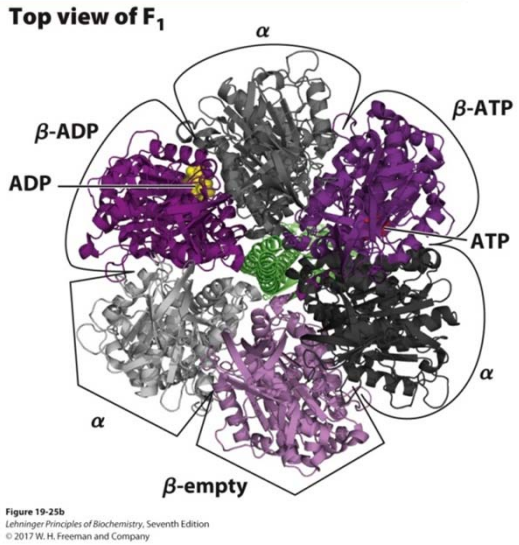


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Coupling Proton Translocation to ATP Synthesis

- Proton translocation causes a rotation of the F_0 subunit and the central shaft γ.
- This causes a conformational change within all the three $\alpha\beta$ pairs.
- The conformational change in one of the three pairs promotes condensation of ADP and P_i into ATP.

Net Production of ATP by Oxidation of Glucose (and Other Fuels) Varies

- In eukaryotic systems, organellar segregation prevents NADH from the cytosol from directly entering the electron-transport chain at Complex I.
  - NAD^+ pools are kept segregated and cannot directly cross the mitochondrial inner membrane.
  - Two methods are used to feed the electrons from NADH from the cytosol into the mitochondria:
    - malate-aspartate shuttle
    - glycerol-3-phosphate shuttle

Malate-Aspartate Shuttle

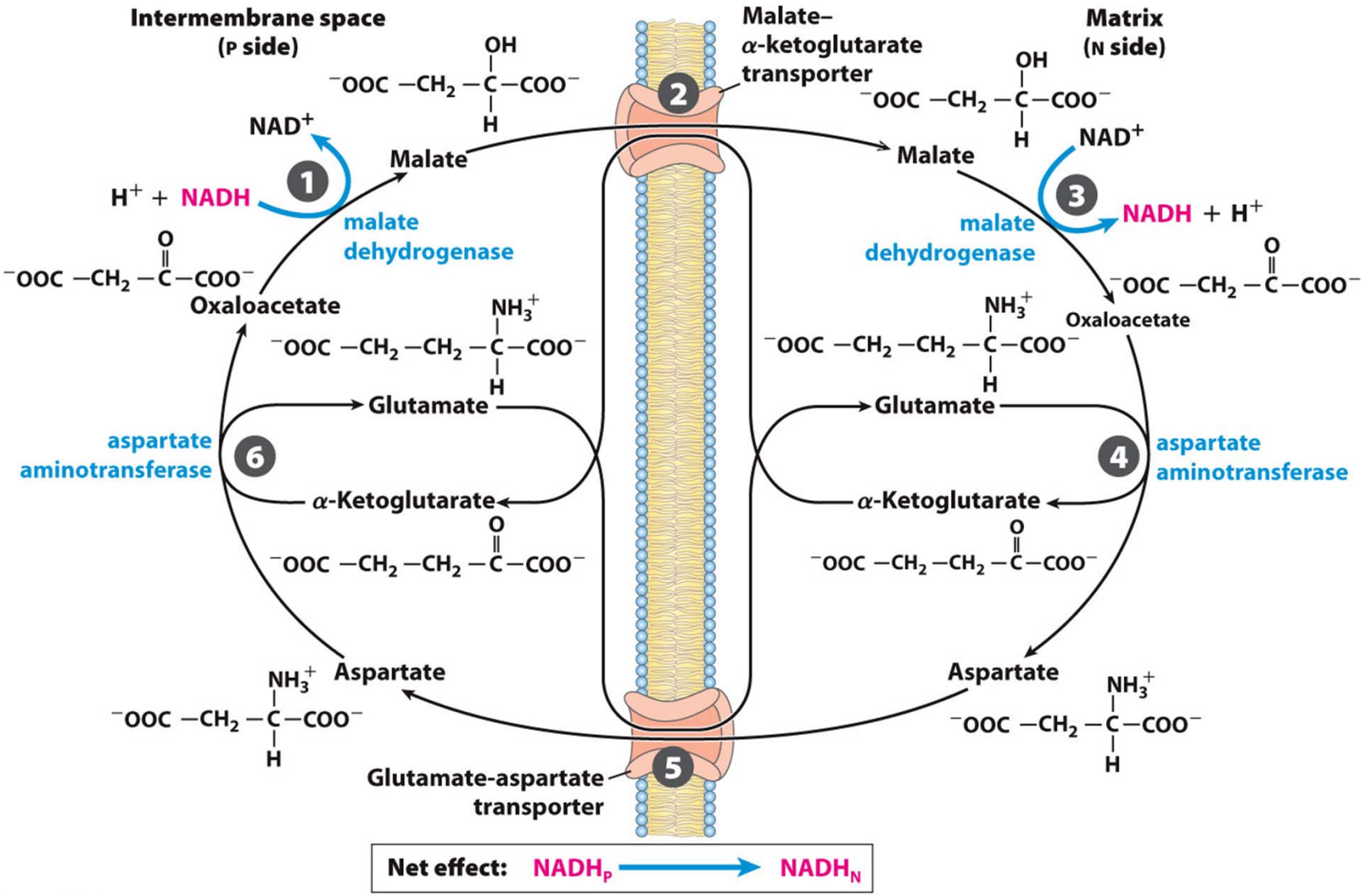


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Glycerol-3-Phosphate Shuttle

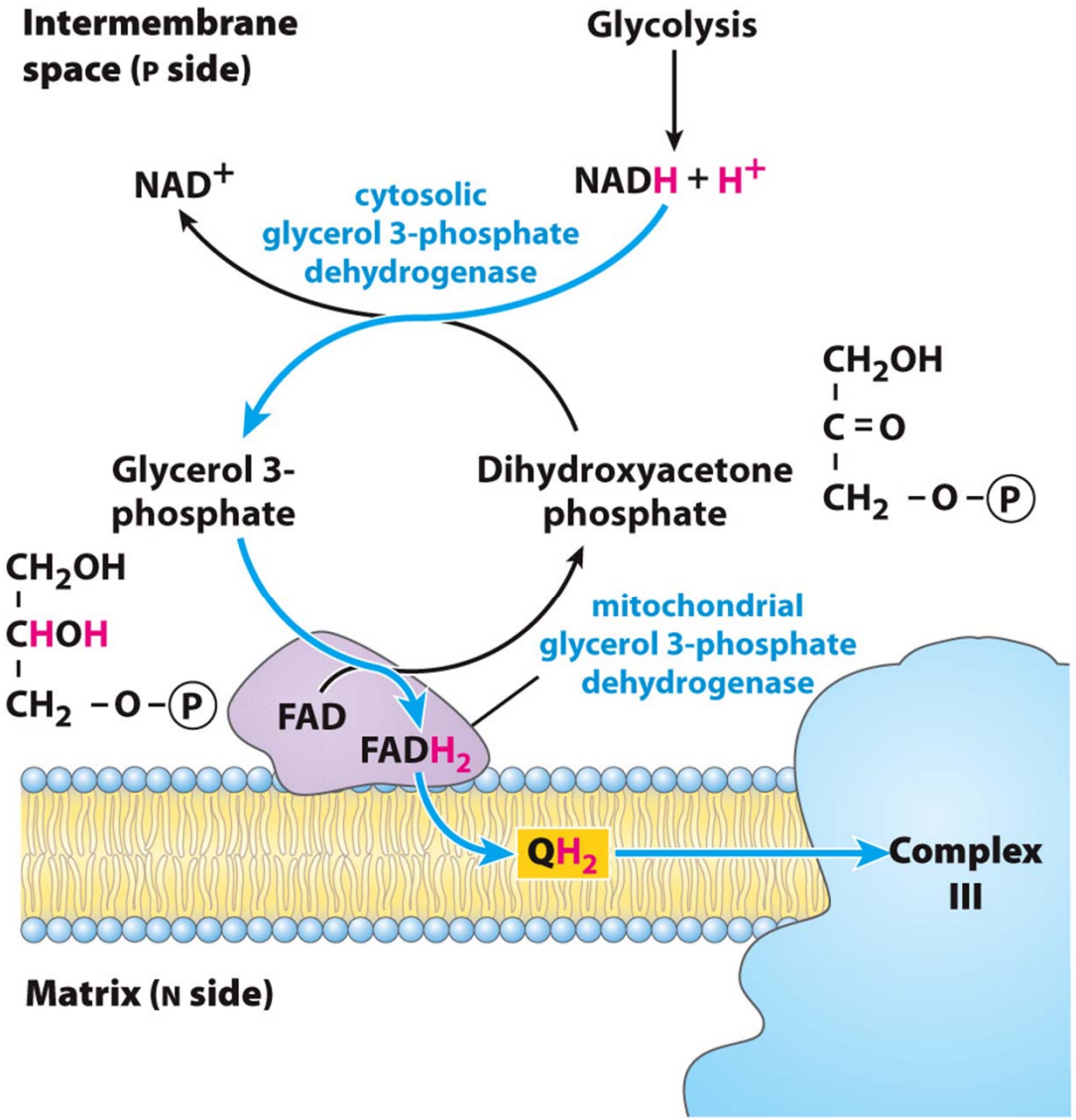


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Net Production of ATP via Catabolic Pathways

TABLE 19-5 ATP Yield from Complete Oxidation of Glucose

Process	Direct product	Final ATP
Glycolysis	2 NADH (cytosolic) 2 ATP	3 or 5a 2
Pyruvate oxidation (two per glucose)	2 NADH (mitochondrial matrix)	5
Acetyl-CoA oxidation in citric acid cycle (two per glucose)	6 NADH (mitochondrial matrix) 2 $FADH_2$ 2 ATP or 2 GTP	15 3 2
Total yield per glucose		30 or 32

[a]If the malate/aspartate shuttle is used to transfer reducing equivalents into the mitochondrion, yield is 5 ATP. If the glycerol 3-phosphate shuttle is used, the yield is 3 ATP.

Regulation of Oxidative Phosphorylation

- Primarily regulated by substrate availability
  - NADH and ADP/P_i
- Inhibitor of F_1 (IF_1)
  - prevents hydrolysis of ATP during low oxygen
  - only active at lower pH, encountered when electron transport it stalled (i.e., low oxygen)
- Inhibition of OxPhos leads to accumulation of NADH.
  - causes feedback inhibition cascade up to PFK-1 in glycoysis

Regulation of Oxidative Phosphorylation

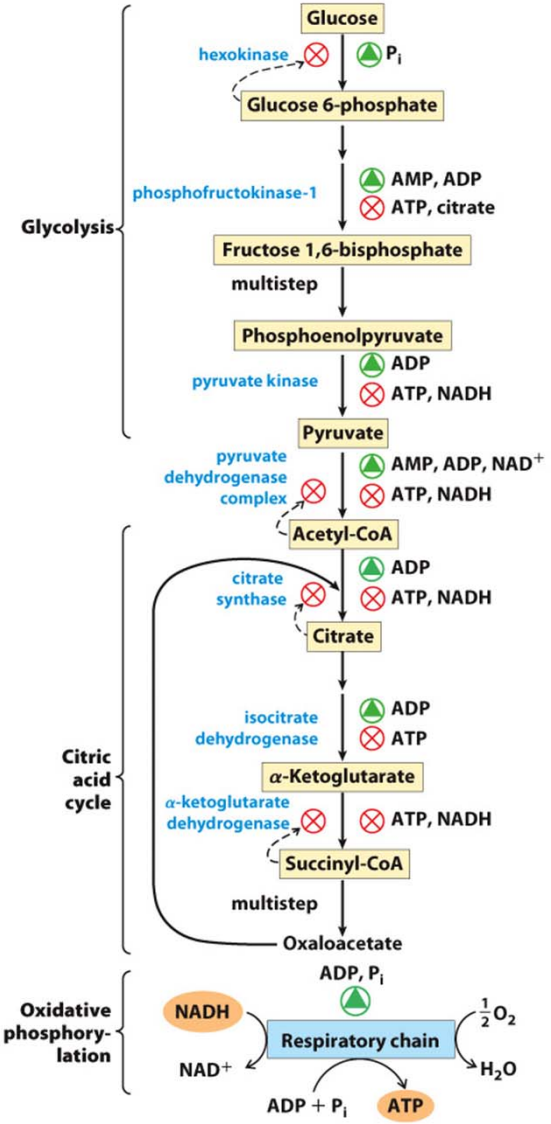


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